SAMY ZAIDI

Mechanical Engineering Student | Mechanical Design, CAD, Simulation and Prototyping

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PROFILE

Bachelor's student in mechanical engineering at Polytechnique Montréal, with two years of experience on the mechanical team of the Esteban solar car and two personal projects taken from requirements definition to detailed design. Skilled in CAD design, mechanical sizing, finite element analysis and automating calculations in Python. Seeking an internship or a position in mechanical design.

EDUCATION

Bachelor of Engineering, Mechanical Engineering | Polytechnique Montréal, Montréal, QC **In progress**
Mechanical design, strength of materials, numerical simulation, manufacturing, thermodynamics

DEC (College Diploma) in Natural Sciences | Collège de Maisonneuve, Montréal, QC **2024**

EXPERIENCE

Mechanical Team Member | Esteban, Polytechnique Montréal solar car **2024 – 2026**

- Contributed to the design and integration of three Esteban 12 components: parking brake (2024), suspension trailing arm (2025) and steering drag rod with its bracket (2026).
- Developed a Python program that automates the calculation of the parking brake's main sizing parameters from manufacturer data.
- Performed the CAD modelling and load analysis of the trailing arm, then evolved its geometry into a part that was manufactured and integrated into the vehicle.
- Collaborated with the mechanical and electrical teams to adapt parts to space, mounting, mass and manufacturing constraints; presented and defended my technical choices to the team.

PERSONAL PROJECTS

Daedalus HES | Passive-assist knee exoskeleton (FreeCAD, Python, CalculiX) **In progress**

- Single-handedly designed a fully parametric knee exoskeleton with compression springs and a ratchet: FreeCAD model generated by Python scripts, with per-part STEP files and an automatic bill of materials.
- Defined four reference sizes (5th-percentile female to 95th-percentile male) from ANSUR II anthropometric data (over 6,000 subjects); 11 to 27 N·m of assistance at 90° of flexion, for 2.7 kg per leg.
- Validated the design: collision checks from −2° to 150°, stress calculations for critical parts (safety factors of 2.2 to 13.8 with a 2× overload) and fatigue sizing of the springs.
- Designed an instrumented test bench (load cell, AS5600 angle sensor) and wrote a two-phase test protocol.

CNC Tripteron | 3-axis parallel-kinematics CNC mill (FreeCAD, Python, 3D printing) **In progress**

- Designed a 200 × 200 × 120 mm CNC mill with three articulated legs, using a parametric skeleton and inverse kinematics to validate reach across the entire work volume.
- Automated the full assembly with Python macros (2040 aluminium frame, actuators, hardware, electronics), including collision detection and a model-generated bill of materials.
- Designed 27 3D-printed parts (PLA+ and PETG) with fit clearances tuned for 608ZZ bearings and LM8UU linear bearings.

SKILLS AND LANGUAGES

Design: 3D CAD and technical drawings, sizing of mechanical components, design for manufacturing and assembly (DFMA), material selection, machine elements, subsystem integration

Analysis and simulation: strength of materials, finite element analysis (FEA), forces, torques and safety factors, fatigue, thermodynamics

Software: FreeCAD, CATIA V5, 3DEXPERIENCE, Simcenter 3D, NX Nastran, CalculiX, EES, LabVIEW, Microsoft Excel

Programming: Python (scientific computing, data analysis, CAD automation), VBA, C++ (ESP32, Arduino)

Prototyping and testing: 3D printing, test bench design, instrumentation and data acquisition, motor control

Languages: French, English, Kabyle